



One step in-situ synthesis of $\text{Na}_3\text{V}_2(\text{PO}_4)_3/\text{Na}_3\text{V}_3(\text{PO}_4)_4$ biphasic coexisted cathode with high energy density by inducing of polyvinylpyrrolidone for sodium ion batteries

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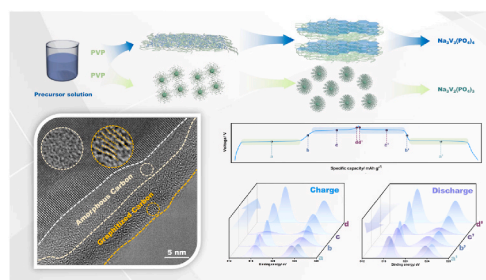
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HIGHLIGHTS

- $\text{Na}_3\text{V}_2(\text{PO}_4)_3/\text{Na}_3\text{V}_3(\text{PO}_4)_4$ hybrid material is synthesized in one-step using PVP.
- The formation mechanism of $\text{Na}_3\text{V}_2(\text{PO}_4)_3$ and $\text{Na}_3\text{V}_3(\text{PO}_4)_4$ induced by PVP is investigated.
- N-doped carbon coating formed by PVP after sintering possesses more defects.
- PVP promotes orderly arrangement of carbon materials to form graphitized carbon.
- PVP-NVP submits impressive capacity and energy density.

GRAPHICAL ABSTRACT



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ABSTRACT

The low voltage platform (~ 3.4 V) and energy density of $\text{Na}_3\text{V}_2(\text{PO}_4)_3$ are far from the requirement of practical applications. Herein, $\text{Na}_3\text{V}_2(\text{PO}_4)_3/\text{Na}_3\text{V}_3(\text{PO}_4)_4$ biphasic coexisted cathode is synthesized by a sol-gel method in one step. It exhibits an extra stable and sustained high voltage plateau (~ 3.9 V) derived from the $\text{Na}_3\text{V}_3(\text{PO}_4)_4$. Polyvinylpyrrolidone (PVP) acts as a structural guiding agent to induce the formation of $\text{Na}_3\text{V}_3(\text{PO}_4)_4$ and behaves as a chelating agent to facilitate the synthesis of $\text{Na}_3\text{V}_2(\text{PO}_4)_3$. Nitrogen atoms on five-membered rings in PVP are reductive to make the reaction of $\text{V}^{5+}/\text{V}^{3+}$ faster and more complete. Furthermore, PVP decomposes into clusters and free radicals to form uniform carbon with relatively ordered structure. The generated nitrogen-doped carbon coating formed by PVP possesses more beneficial defects to accelerate the transport of electrons, resulting in superior kinetic characteristics. Ex-situ X-ray photoelectron spectroscopy demonstrates the reversible $\text{V}^{3+}/\text{V}^{4+}$ redox pair features both at voltage of 3.4 and 3.9 V, further indicating the biphasic coexistence of $\text{Na}_3\text{V}_2(\text{PO}_4)_3/\text{Na}_3\text{V}_3(\text{PO}_4)_4$. Correspondingly, $\text{Na}_3\text{V}_2(\text{PO}_4)_3/\text{Na}_3\text{V}_3(\text{PO}_4)_4$ biphasic material submits a capacity of $119.17 \text{ mAh g}^{-1}$ at 0.1C and delivers an impressive energy density of $471.19 \text{ Wh kg}^{-1}$. This novel biphasic cathode significantly improves the energy density of $\text{Na}_3\text{V}_2(\text{PO}_4)_3$ to promote its practical application.

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1. Introduction

With the development of new energy technology and the explosive growth of portable intelligent devices, the global demand for energy storage batteries is growing rapidly [1–4]. Although lithium-ion batteries (LIBs) is widely used because of its excellent energy storage performance, it is limited by its lack of resources and high cost [5–7]. As sodium element and lithium element get along with the same main group of elements, have similar physical and chemical properties, so sodium-ion batteries (SIBs) is considered as the most promising energy storage battery [8–11]. Both SIBs and LIBs are rocking chair batteries (RCB). Therefore, the cathode material of SIBs needs to have an open stable channel to ensure that Na^+ can be quickly embedded and de-embedded when the battery is charged and discharged [12–14].

$\text{Na}_3\text{V}_2(\text{PO}_4)_3$ (NVP) is considered to be an ideal cathode material for sodium ion batteries due to its open and stable 3D skeleton [15–17]. Benefiting from the NASICON framework and two extractable Na^+ , the theoretical reversible specific capacity of NVP is up to 117.6 mAh g^{-1} [18,19]. However, due to the low intrinsic conductivity and ionic conductivity of NVP materials, the energy density and discharge specific capacity of NVP are far lower than the theoretical value in practical application, which greatly limits its commercial application [9,20–22]. To overcome this problem, numerous methods has been explored, including carbon cladding or recombination [8,23–26], ion doping [21, 27–29] and multiphase system doping [25,30,31]. Regulating the NVP synthesis process by adding excessive special carbon sources is a promising way to improve the performance of NVP. Zhang et al. [32] added polyethylene glycol, a polymer with low melting point, to melt infiltration of NVP precursor and form a honeycomb carbon network in the synthesis process of NVP, which inhibited the agglomeration of NVP particles and formed an interconnected conductive carbon network. Xiong et al. [8] developed a simple polymer-stabilized drop template strategy for the synthesis of porous single-crystal NVP. By adjusting the size of the droplet template, the pore structure, pore size and specific surface area of the sample can be easily controlled.

Herein, we use the linear polymer polyvinylpyrrolidone (PVP) as the structural-guiding agent to induce the formation of $\text{Na}_3\text{V}_2(\text{PO}_4)_3$ and $\text{Na}_3\text{V}_3(\text{PO}_4)_4$. PVP can effectively avoid agglomeration of particles and improve the crystallinity of the two-phase materials. $\text{Na}_3\text{V}_3(\text{PO}_4)_4$ has the highest operating voltage ($\sim 3.9 \text{ V}$) and excellent structural stability among the sodium-containing vanadium orthophosphates currently known. However, it is limited by its low theoretical capacity (45.00 mAh g^{-1}). The formation in-situ of $\text{Na}_3\text{V}_3(\text{PO}_4)_4$ significantly improves the energy density of the whole material without significantly affecting the specific capacity of the two-phase material. In addition, PVP acts as an additional carbon source to carbon coating the two-phase materials, which effectively improves its electronic conductivity and ionic conductivity. Therefore, $\text{Na}_3\text{V}_2(\text{PO}_4)_3/\text{Na}_3\text{V}_3(\text{PO}_4)_4$ exhibits excellent electrochemical properties. Notably, the optimized sample showed the amazing electrochemical performance: It reveals the reversible specific capacity of $119.17 \text{ mAh g}^{-1}$ at 0.1 C and possesses an energy density of $471.19 \text{ Wh kg}^{-1}$ at 0.1 C . It also displays an average energy density of $408.12 \text{ Wh kg}^{-1}$ after 200 cycles at 0.1 C , corresponding to a high retention of 86.61%. The one-step synthesis of two-phase materials showed excellent electrochemical properties and provided a new model

for future research.

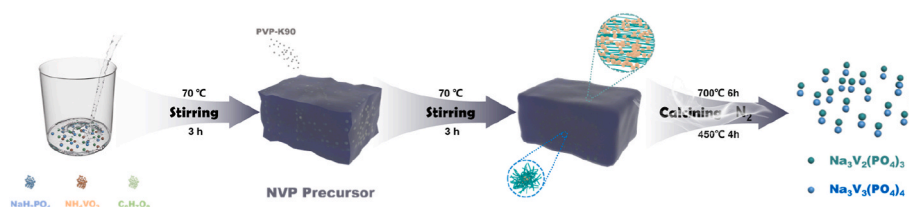
2. Experiment

The optimized sample was synthesized by sol-gel method in one step and the entire synthesis process is shown in Scheme 1. The synthesized optimized sample was abbreviated as PVP-NVP. The pristine NVP/C sample was synthesized by the same route without PVP and denoted as NVP/C. The details of sample preparation, battery assembling, corresponding characterization and electrochemical tests are presented in the Supporting Information.

3. Results and discussions

To explore the phase composition and crystallinity of two-phase materials synthesized by one-step method, The sample were tested by X-ray diffraction. The XRD patterns of the samples are displayed in Fig. 1 and Fig. S1. Fig. 1a shows the XRD patterns of PVP-NVP and NVP/C. In the two-phase sample PVP-NVP, sharp X-ray diffraction peaks at 14.30° , 20.17° , 23.84° and 32.05° can be intuitively observed, which can be undoubtedly attributed to NASICON-type $\text{Na}_3\text{V}_2(\text{PO}_4)_3$ with a space group of R-3c. The semi-quantitative analysis of the refined results by Highscore software shows that the mass ratios of $\text{Na}_3\text{V}_2(\text{PO}_4)_3$ and $\text{Na}_3\text{V}_3(\text{PO}_4)_4$ in PVP-NVP is 79:21. Furthermore, different masses of PVP were added to the precursor and the molar ratio of $\text{Na}_3\text{V}_2(\text{PO}_4)_3/\text{Na}_3\text{V}_3(\text{PO}_4)_4$ was calculated. Specifically, when the mass of PVP was 1, 2 and 4 g in the precursor, the molar ratios of $\text{Na}_3\text{V}_2(\text{PO}_4)_3/\text{Na}_3\text{V}_3(\text{PO}_4)_4$ was 71:29, 73:27 and 80:20, respectively (Figs. S1b–d). Fig. 1b displays the refined XRD patterns of PVP-NVP, and the low R value indicates the reliability of the refined results. The sharp characteristic peaks testify to the high crystallinity of PVP-NVP, which can be attributed to the regulatory role of PVP in the synthesis of $\text{Na}_3\text{V}_2(\text{PO}_4)_3$ and $\text{Na}_3\text{V}_3(\text{PO}_4)_4$. On the one hand, PVP, as $\text{Na}_3\text{V}_2(\text{PO}_4)_3$ chelating agent, effectively increases the steric hindrance between particles, thus preventing agglomeration and making $\text{Na}_3\text{V}_2(\text{PO}_4)_3$ -phase materials with small grain size. On the other hand, during the growth of $\text{Na}_3\text{V}_3(\text{PO}_4)_4$, PVP mainly acts as a one-dimensional cross-linked organic template to induce precursor growth into $\text{Na}_3\text{V}_3(\text{PO}_4)_4$ -phase with layered structures. Specifically, the amide structure in PVP allows the precursor to be attracted to the surface of the cross-linked template through coordination and induces the formation of layered $\text{Na}_3\text{V}_3(\text{PO}_4)_4$ crystals.

In order to determine the components of PVP-NVP more intuitively, the characteristic patterns of $\text{Na}_3\text{V}_2(\text{PO}_4)_3$ and $\text{Na}_3\text{V}_3(\text{PO}_4)_4$ are compared with the XRD pattern of PVP-NVP, which are shown in Fig. 1c. The main X-ray diffraction peaks at 14.30° , 20.07° , 20.33° , 31.56° , 32.02° and 35.60° are good matching with $\text{Na}_3\text{V}_2(\text{PO}_4)_3$ (ICSD No. 248140). The sharp peaks at 9.02° , 14.58° , 22.53° , 26.67° and 29.41° correspond to the (2 0 0), (1 1 0), (5 1 0), (1 1 2) and (2 2 0) crystal planes of $\text{Na}_3\text{V}_3(\text{PO}_4)_4$ (ICSD No. 95532) respectively. Furthermore, the impurity peaks at 9.68° , 12.43° , 13.45° , 15.21° , 25.39° and 29.81° were assigned to C_3N_4 and C_2N_6 compounds formed by sintering of PVP, which is beneficial to improve the conductivity of the composite [33,34]. It is worth noting that the characteristic peaks representing $\text{Na}_3\text{V}_2(\text{PO}_4)_3$ and $\text{Na}_3\text{V}_3(\text{PO}_4)_4$ in the composite are exceedingly obvious, which indicates that the crystallinity of both phases is high in



Scheme 1. Flow diagram of the synthesis about the PVP-NVP samples.

the composite. In addition, all characteristic peaks can be attributed to $\text{Na}_3\text{V}_2(\text{PO}_4)_3$ and $\text{Na}_3\text{V}_3(\text{PO}_4)_4$ without additional impurity peaks, which undoubtedly proves the successful synthesis of $\text{Na}_3\text{V}_2(\text{PO}_4)_3$ – $\text{Na}_3\text{V}_3(\text{PO}_4)_4$ two-phase electrodes.

Fig. 1d shows ^{23}Na solid-state nuclear magnetic resonance (SSNMR) spectra of PVP-NVP and NVP/C respectively. There were two kinds of Na in the NVP/C samples, corresponding to signal peaks at 74 and -65 ppm in the ^{23}Na SSNMR spectrum. One of them is the sodium containing site (6b) located in octahedral position in the hexagon coordination environment, denoted as the M1 site (Na1). The other is the octahedral sodium containing site (18e), denoted as M2 site (Na2). Fig. 1e visually shows the positions of sodium atoms in $\text{Na}_3\text{V}_2(\text{PO}_4)_3$ in two different environments. There were three kinds of Na in PVP-NVP samples. Among them, signal peaks at 7 and -8 ppm were classified as M2 and M1 sites in NVP, respectively. Compared with NVP/C, the signal peaks of PVP-NVP are sharper and easier to distinguish, which can be attributed to the higher crystallinity of PVP-NVP, and it corresponds to the curves of XRD spectrum. The signal peak of PVP-NVP is sharper and the half-peak width is lower, which can be attributed to that the fast ionic motion can average dipole–dipole interaction, thus contributing to a narrow line-width of NMR signal. Specifically, the presence of $\text{Na}_3\text{V}_2(\text{PO}_4)_3$ and $\text{Na}_3\text{V}_3(\text{PO}_4)_4$ two-phase material in PVP-NVP results in the appearance of the lattice distortion. This promotes the fast movement of sodium ions, averaging the dipole–dipole interaction. Furthermore, NVP/C crystal has a high degree of agglomeration and low crystallinity, presenting anisotropy. The interaction of anisotropy leads to the interaction of spatial structure, thus increasing the half-peak width of signal peak. For PVP-NVP, PVP acts as a template and structural guide to make

$\text{Na}_3\text{V}_3(\text{PO}_4)_4$ directional growth, reducing the degree of NVP agglomeration to achieve uniform distribution of multi-directional disordered crystals with apparent isotropic appearance. In addition, the coordination environment of Na atom is changed by the presence of lattice distortion between $\text{Na}_3\text{V}_2(\text{PO}_4)_3$ and $\text{Na}_3\text{V}_3(\text{PO}_4)_4$ two-phase in PVP-NVP, resulting in that the spectrum peak approaches 0 ppm. Notably, the additional signal peak at -45 ppm can be attributed to the sodium site of monoclinic $\text{Na}_3\text{V}_3(\text{PO}_4)_4$ which is displayed in Fig. 1e, demonstrating the coexistence of layered $\text{Na}_3\text{V}_3(\text{PO}_4)_4$ and irregular $\text{Na}_3\text{V}_2(\text{PO}_4)_3$ two-phase materials.

The microstructure of PVP-NVP was characterized by scanning electron microscopy (SEM), as shown in Fig. 2. Intuitively, PVP-NVP delivers small particle size, low agglomeration and uniform dispersion (Fig. 2a). This can be attributed to the PVP, as a chelating agent, effectively improves the steric resistance between NVP precursor particles, thus effectively avoiding agglomeration in the sintering process. Fig. 2b shows in detail the morphology of PVP-NVP two-phase materials at high magnification. Specifically, the sample presented disordered distribution and layered superposition. Fig. 2c shows the EDS image of PVP-NVP. Obviously, the corresponding elements of C, N, V, Na, P and O are all detected. It is worth noting that the distribution of N element is quite different from that of C element, which can be attributed to the decomposition of PVP into small molecule and cluster forms during sintering. In addition, the wide distribution of carbon elements can be attributed to the presence of carbon substrates and carbon coatings during the synthesis process, which contribute to rapid electronic transport.

Further, transmission electron microscope (TEM) and high-

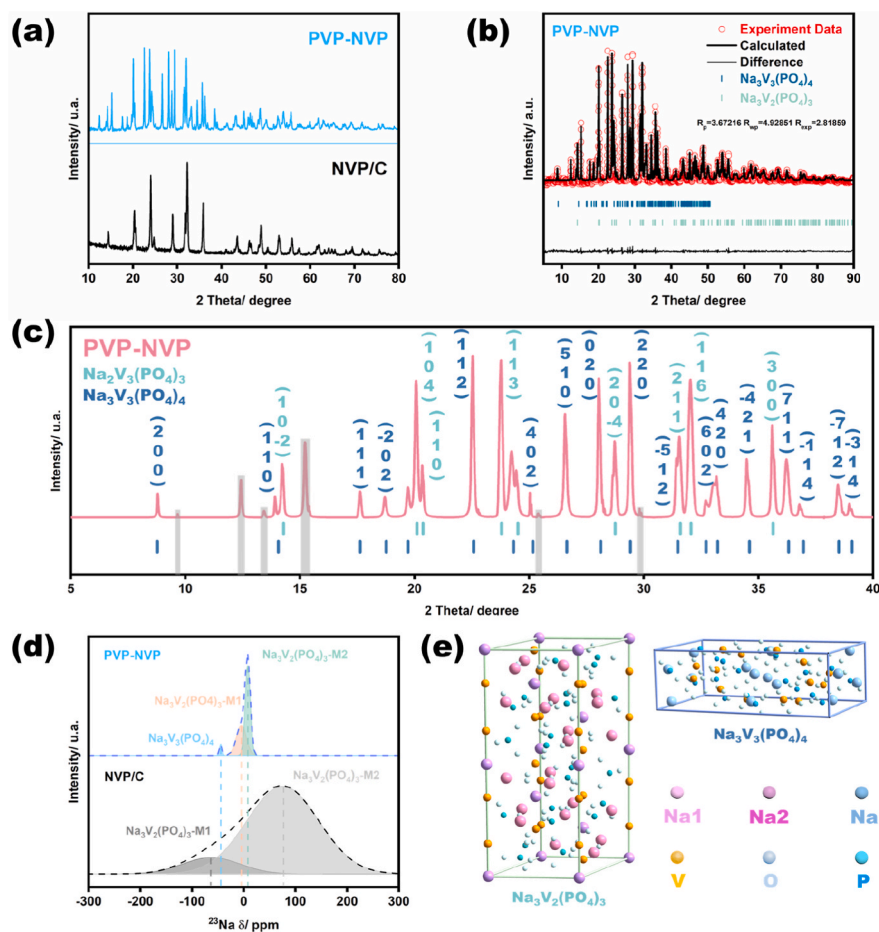


Fig. 1. (a) XRD curves of PVP-NVP and NVP/C. (b) The Rietveld refined XRD patterns of PVP-NVP. (c) Comparison of the XRD patterns of PVP-NVP and the characteristic spectra of $\text{Na}_3\text{V}_2(\text{PO}_4)_3$ / $\text{Na}_3\text{V}_3(\text{PO}_4)_4$. (d) ^{23}Na solid-state nuclear magnetic resonance (SSNMR) spectra of the PVP-NVP and NVP/C. (e) Crystal structure of $\text{Na}_3\text{V}_2(\text{PO}_4)_3$ and $\text{Na}_3\text{V}_3(\text{PO}_4)_4$.

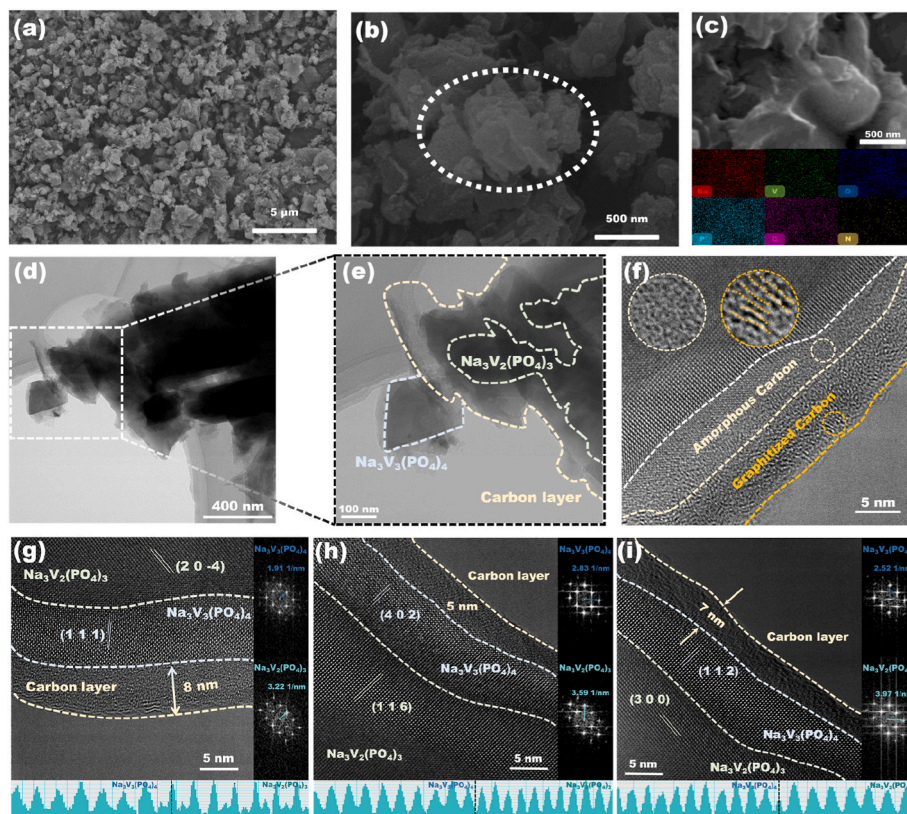


Fig. 2. (a) SEM images of PVP-NVP. (b) High magnification SEM image of PVP-NVP. (c) EDS mapping images of PVP-NVP. (d–i) TEM/HRTEM images of PVP-NVP.

resolution transmission electron microscope (HRTEM) images of the PVP-NVP composites are shown in Fig. 2. Fig. 2d and e clearly shows the composition materials of PVP-NVP, namely irregular $\text{Na}_3\text{V}_2(\text{PO}_4)_3$ and lamellar $\text{Na}_3\text{V}_3(\text{PO}_4)_4$, and carbon coating. Remarkably, PVP not only induced the synthesis of two-phase materials during the synthesis process, but also formed a highly ordered carbon coating after sintering which can be observed by Fig. 2f. As shown in Fig. 2g, the existence of two kinds of carbon can be clearly observed. The appearance of highly graphitized carbon layer can be attributed to the decomposition of PVP into clusters and free radicals during carbonization, resulting in the formation of homogeneous carbon nanoparticles with relatively ordered structure, and produces highly graphitized carbon coating. The highly ordered carbon coating can effectively improve the electronic conductivity of the composite material and allow the rapid transfer of electrons on the surface of the material. From Fig. 2g–i, two phases of the composite, including highly crystalline $\text{Na}_3\text{V}_2(\text{PO}_4)_3$ and $\text{Na}_3\text{V}_3(\text{PO}_4)_4$, can be visually detected, which is consistent with X-ray diffraction results. There are obvious lattice fringes in Fig. 2g, h and i, corresponding to the $\text{Na}_3\text{V}_2(\text{PO}_4)_3$ phase and $\text{Na}_3\text{V}_3(\text{PO}_4)_4$ phase respectively. Specifically, the lattice fringe spacing observed in the HRTEM images which ascribe to the (1 1 1), (4 0 2) and (1 1 2) crystal planes of $\text{Na}_3\text{V}_3(\text{PO}_4)_4$ correspond to the characteristic peaks of 17.61° , 25.03° and 22.53° in the XRD patterns (Fig. 1c). And the lattice fringe spacing observed in the HRTEM images which ascribe to the (2 0 4), (1 1 6) and (3 0 0) crystal planes of $\text{Na}_3\text{V}_2(\text{PO}_4)_3$ correspond to the characteristic peaks of 28.75° , 32.04° and 35.61° in the XRD patterns (Fig. 1c). After Fast Fourier Transform (FFT) of the calibrated lattice stripes in the HRTEM image, multiple pairs of points are obtained, as shown in the inset of Fig. 2g–i. The crystal plane spacing of the corresponding crystal plane is obtained by measuring the distance from the point to the center and calculating the inverse. This corresponds to the direct measurement of the crystal plane spacing, proving the existence of two phases, $\text{Na}_3\text{V}_2(\text{PO}_4)_3$ and $\text{Na}_3\text{V}_3(\text{PO}_4)_4$. Notably, as shown in Fig. S2, disorder and lattice

distortion can be observed at the $\text{Na}_3\text{V}_2(\text{PO}_4)_3/\text{Na}_3\text{V}_3(\text{PO}_4)_4$ phase boundary. FFT and Inverse Fast Fourier Transform (IFFT) are performed on the part where the lattice distortion is presented, and the clear images of the lattice distortion can be obtained, as shown in the insets of Figs. S2a and b. Obviously, the patterns of $\text{Na}_3\text{V}_2(\text{PO}_4)_3$ and $\text{Na}_3\text{V}_3(\text{PO}_4)_4$ can be easily distinguished. In addition, the lattice distortion can generate abundant external defects and active sites for ion transport [30,35].

Fig. 3a shows the TGA test results for all samples. The carbon contents of NVP/C and PVP-NVP are 4.57 and 8.42 wt%, respectively. The carbon content of PVP-NVP was higher than that of NVP/C, which can be attributed to the addition of PVP as an additional carbon source. This facilitates the formation of carbon cladding, which creates a highly conductive network that allows electrons to move quickly. Moreover, the TG curves of all samples show an inflection point at 530°C , which can be attributed to the oxidation process of V^{3+} to V^{4+} . In order to further investigate the properties of unique carbon coating which formed by PVP, the Raman spectrum of the sample is analyzed, as shown in Fig. 3b and S3. Raman spectra display two distinct peaks, located near 1343.23 cm^{-1} and 1564.67 cm^{-1} respectively, which can undoubtedly be attributed to the characteristic for defects and disorders of carbon (D-band) and typical graphitized carbon (G-band) of carbon. To acquire the high intensity of Raman test results, the peak intensity ratio (I_D/I_G) of the sample is determined by testing repeatedly three times (Fig. S3). Comparing with pristine NVP/C (0.97), the average peak intensity ratio (I_D/I_G) of PVP-NVP is much higher (1.04) [36], which can be attributed to the N-doped carbon coating formed after sintering by PVP as a carbon source, significantly improving the disorder degree of the carbon layer. The nitrogen-doped carbon layer effectively improves the defect degree of the carbon layer, so that the carbon coating layer not only fast conducts electricity, but also enables the rapid diffusion of Na^+ [36–38].

In order to further investigate the doping behavior of nitrogen in carbon layer, X-ray photoelectron spectroscopy (XPS) was performed on

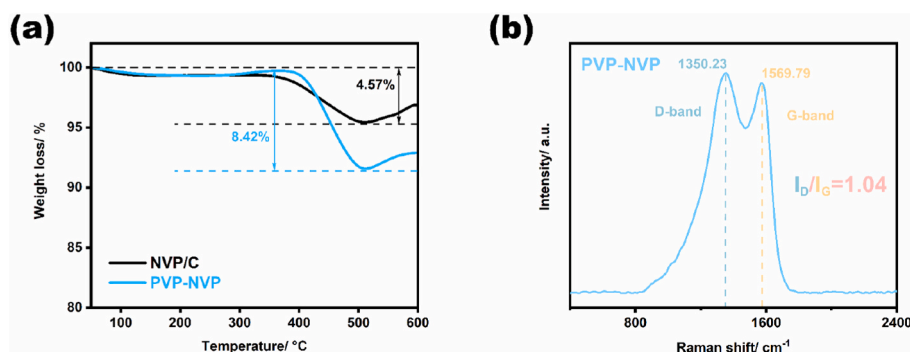


Fig. 3. (a) The TGA curves of NVP/C and PVP-NVP. (b) Raman spectra of PVP-NVP.

PVP-NVP samples. Fig. 4a shows two peaks at 516.25 eV and 523.45 eV can be attributed to the excitations of $V^{3+} 2p_{3/2}$ and $V^{3+} 2p_{1/2}$ in PVP-NVP, respectively. This indicates the presence of V^{3+} . And the peak at 520.35 eV in Fig. 4a belongs to O 1s. The peak at 1070.85 eV in Fig. 4b can be attributed to Na 1s, which proves the stable existence of Na element in both crystal structures. Intuitively, O element mainly exists in the form of O-P and C-O, and the peak at 535.30 eV is the satellite peak of O element (Fig. 4c). Fig. 4d displays several prominent peaks with binding energies of 283.95, 285.15, and 288.15 eV, corresponding to C-C, C-N, and C-O bonds, respectively [39,40], which can reasonably explain the increase of carbon layer disorder in Raman spectrum (Fig. 3b). In addition, C-C bond can significantly enhance the conductivity of free π bond electrons, which is beneficial to improve the rate properties of materials [41]. Fig. 4e shows the existing form of nitrogen in the carbon layer. Nitrogen mainly exists in the form of pyridinic nitrogen (397.85 eV), pyrrolic nitrogen (399.80 eV), quaternary nitrogen (402.50 eV) and N-O band (397.50 eV). All of these nitrogen species can improve the electronic conductivity of the carbon coating. In addition, both pyridine nitrogen and pyrrole nitrogen can produce a large number of external defects and active sites. These defects and active sites contribute to the rapid diffusion of sodium ions when they are inserted/de-inserted into the carbon layer during charge and discharge. Therefore, nitrogen doped carbon layer will effectively improve the electron and ion transport capacity of the carbon layer.

The Brunauer-Emmett-Teller (BET) surface area and accurate pore distribution of PVP-NVP and pristine NVP/C were obtained by N_2

adsorption/desorption test, as shown in Fig. 4f and Fig. S3. BET curves of all samples can be classified as type IV isotherms, which indicates that micropores and mesoporous pores mainly exist on the surface of materials. Most of the pore sizes in PVP-NVP composites are 3.39 nm, with an average pore size of 11.17 nm. What's more, PVP-NVP delivers a surface area of $20.47 \text{ m}^2 \text{ g}^{-1}$, which much higher than that of NVP/C ($10.90 \text{ m}^2 \text{ g}^{-1}$). This can be attributed to the exist of NVP of layered structure. The high specific surface area enables the electrode to fully contact with the electrolyte, providing more active sites for the transmission of Na^+ between the electrode and the electrolyte.

The electrochemical properties of all composites were tested in the voltage range from 2.3 to 4.1 V and the data of NVP/C and PVP-NVP are presented in Fig. 5. Besides, the electrochemical properties of $Na_3V_3(PO_4)_4$ are shown in Fig. S4. It can be intuitively observed from Fig. 5a that the reversible discharge specific capacities of optimized sample are superior to NVP/C. It is worth noting that the composites prepared by PVP addition all have an obvious second discharge platform at 3.9 V, which can be attributed to the second phase substance $Na_3V_3(PO_4)_4$ in PVP-NVP. Specifically, the GCD curve of PVP-NVP electrode has a stable voltage platform and low polarization voltage, which is basically close to the ideal state. It releases the highest initial capacity of $119.17 \text{ mAh g}^{-1}$, far more than pristine NVP/C (67.40 mAh g^{-1}). Besides, as depicted in Fig. 5b, the pristine NVP/C submits the discharge capacity of 67.40 mAh g^{-1} at first cycle and only remains 56.00 mAh g^{-1} after 200 cycles. In contrast, PVP-NVP provides the reversible capacity of $119.17 \text{ mAh g}^{-1}$ and still remains $102.60 \text{ mAh g}^{-1}$

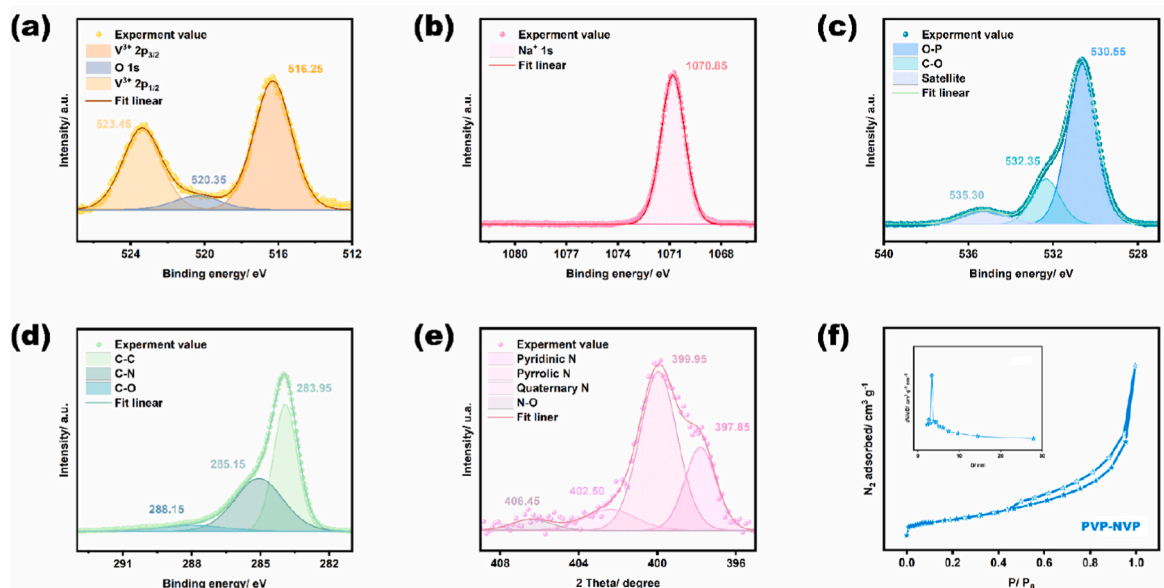


Fig. 4. XPS survey spectrum: V 2p (a), Na 1s(b), O 1s (c), C 1s (d), N 1s (e) spectra of PVP-NVP and Specific area/pore size by BET (f).

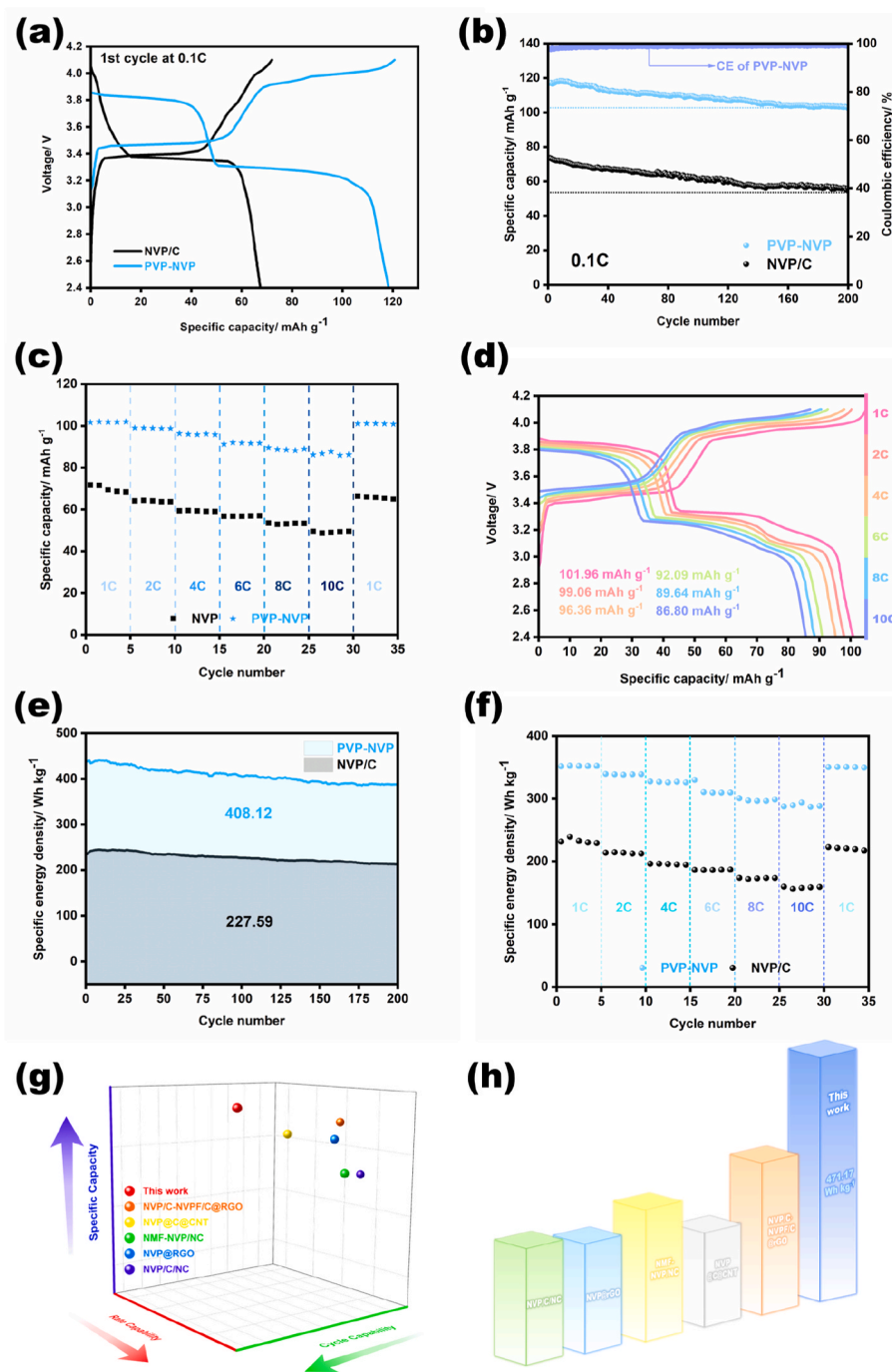


Fig. 5. (a) The initial GCD profiles of NVP and PVP-NVP. (b) The cycle capacity of NVP and PVP-NVP. (c) The rate capacity of all samples. (d) The galvanostatic voltage profiles under different rates for PVP-NVP. Specific energy capabilities of PVP-NVP and NVP/C in circulation (e) and at different magnification rates (f). The comprehensive performance (g) and energy density (h) comparison of PVP-NVP with other previously report.

¹ after 200 cycles, corresponding to a high retention of 89.07%. Furthermore, the coulombic efficiency of PVP-NVP is kept at 100% throughout the cycle. In addition, Fig. 5 exhibits the rate capability of PVP-NVP and NVP/C. Distinctively, PVP-NVP delivers an average capacity of 101.96, 99.06, 96.36, 92.09, 89.64, 86.80, and 101.20 mA h g⁻¹ at the various rates of 1, 2, 4, 6, 8, 10 and 1C. It is obvious that the decreasing trend of PVP-NVP is relatively small as the current density increases. Fig. 5d displays the GCD curves of PVP-NVP electrodes at different current densities. The PVP-NVP electrode has a stable voltage platform of 3.9 V at different current densities, which proves the stability of the Na₃V₂(PO₄)₃/Na₃V₃(PO₄)₄ two-phase electrode and its excellent high magnification performance. What's more, PVP-NVP

displays an amazing high energy density. It possesses an energy density of 471.19 Wh kg⁻¹ at 0.1C. Moreover, PVP-NVP displays an average energy density of 408.12 Wh kg⁻¹ in 200 cycles at 0.1C, almost twice that of NVP/C (227.59 Wh kg⁻¹). Fig. 5f shows the variation trend of energy density of PVP-NVP and NVP/C at different current densities. Distinctly, PVP-NVP exhibits superior energy density at different current densities, much better than NVP/C. In addition, the energy density of PVP-NVP decreases slightly with the increase of current density, proving its superior high-rate performance, which is consistent with Fig. 5c. Furthermore, Fig. 5g compares the comprehensive properties of PVP-NVP and other NVP materials which reported in recent years from the aspects of discharge specific capacity, cycling performance and

multiplier performance. Fig. 5h compares the energy density of PVP-NVP with other NVP materials reported in recent years. Obviously, PVP-NVP electrode possesses excellent electrochemical performance, which can be attributed to the high discharge platform of the two-phase material which induced by PVP.

For purpose of exploring the electrochemical dynamic characteristics of the optimized sample, cyclic voltammetry (CV) test was conducted on them, and the test results are shown in Fig. 6. Fig. 6a shows the CV curves of NVP/C and PVP-NVP in the 2.3–4.1 (V vs. Na/Na⁺) range at a scan speed of 0.1 mV s⁻¹. It can be observed that PVP-NVP exhibits two redox peaks, which are about 3.4 V and 3.9 V and corresponds well to the V³⁺/V⁴⁺ redox pair. The peak at 3.9V has a double peak, which can be attributed to the two-step embedding of Na⁺. The appearance of the

peak at 3.9 V can be attributed to Na₃V₃(PO₄)₄, which proves the successful synthesis of Na₃V₃(PO₄)₄ and is consistent with the observations of XRD and TEM images. The charge-discharge platform of 3.9V significantly increases the energy density of the composite.

In order to further study the electrochemical kinetic characteristics of PVP-NVP composites, cyclic voltammetry tests were carried out at different sweep rates from 0.1 to 0.5 mV s⁻¹ and the results are shown in Fig. 6b. As the scanning speed increases, the anode peak at 3.4 V splits into two, which can be attributed to the presence of two different sodium ion storage sites in the Na₃V₂(PO₄)₃. Even at a scan rate of 0.5 mV s⁻¹, the polarization of the sample is little and still has excellent electrochemical reversibility, which can be seen from the position of the two redox peaks. There is a linear relationship between the peak currents (I_p)

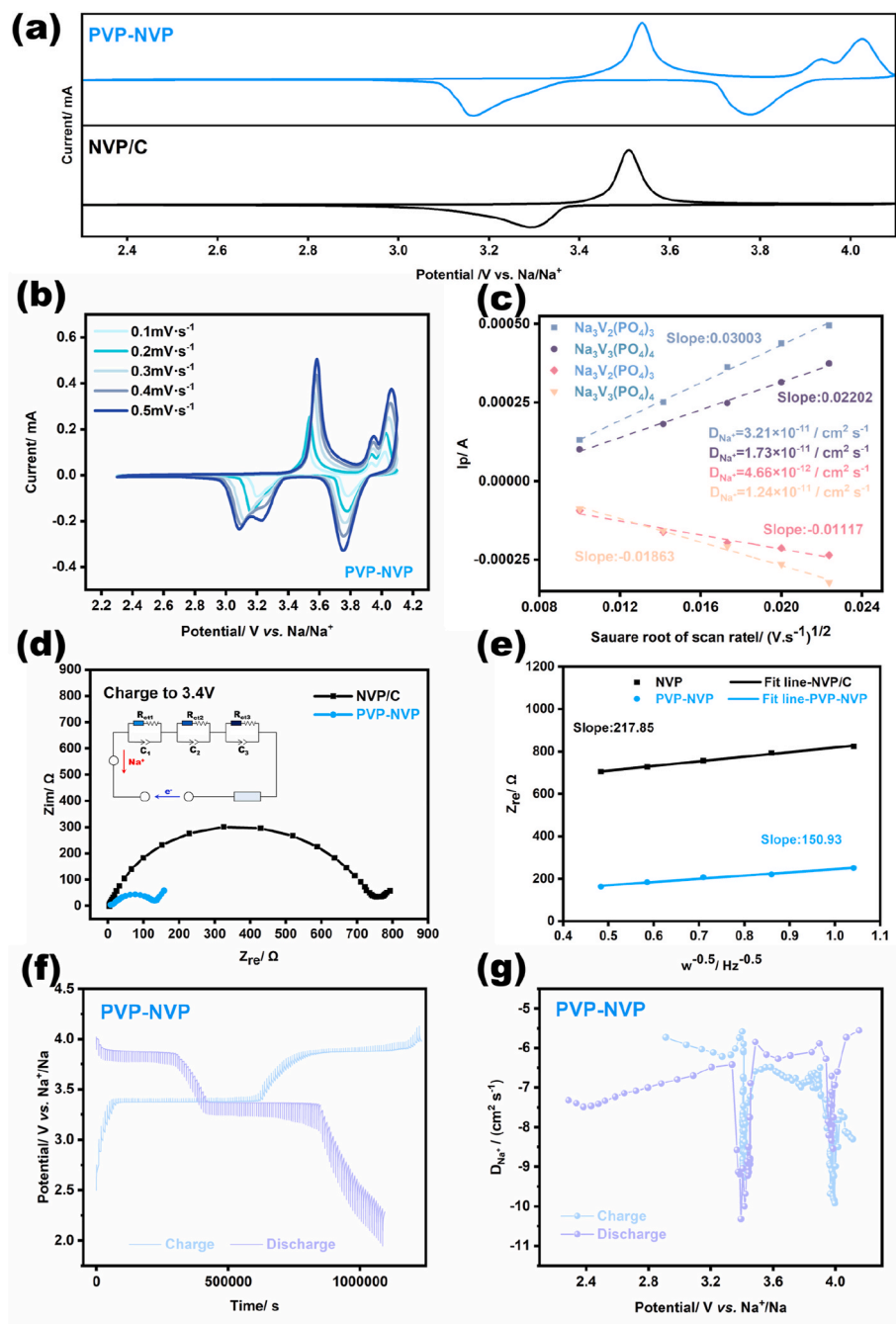


Fig. 6. (a) CV curves of PVP-NVP and NVP/C at 0.1 mV s⁻¹. (b) The CV curves at various scan rates of PVP-NVP. (c) 'I_p vs. v^{1/2}' curve of PVP-NVP. (d) The Nyquist plots of PVP-NVP and NVP/C with equivalent circuit model insert. (e) The linear relationship of 'Z_{re}-ω^{-0.5}' for NVP/C and PVP-NVP. (f) The GITT curves of PVP-NVP. (g) The calculated D_{Na⁺} during charge/discharge process of PVP-NVP.

and the square root of scanning rate ($v^{1/2}$) at different scanning rates of PVP-NVP, and the fitting line is shown in Fig. 6c. The diffusion coefficient of Na^+ (D_{Na^+}) was calculated by the Randles-Sevcik equation.

$$I_p = 2.69 \times 10^5 n^{3/2} A C D^{1/2} v^{1/2} \quad (1)$$

The parameters n , A , C , D and v in Eq. (1) represent the number of exchanged electrons, the effective area of the electrode, the concentration of Na^+ in the electrode, the diffusion coefficient of Na^+ and the scanning rate, respectively. It is worth noting that $\text{Na}_3\text{V}_2(\text{PO}_4)_3$ and $\text{Na}_3\text{V}_3(\text{PO}_4)_4$ have the same n and A values in the calculation, which are 2 and 2.0096 cm^2 , respectively. Table 1 summarizes the D_{Na^+} of NVP/C and PVP-NVP. The D_{Na^+} of PVP-NVP in 3.4V region was $3.21 \times 10^{-11} \text{ cm}^2 \text{ s}^{-1}$ in anode reaction, $4.66 \times 10^{-12} \text{ cm}^2 \text{ s}^{-1}$ in cathode reaction, and 1.73×10^{-11} and $1.24 \times 10^{-11} \text{ cm}^2 \text{ s}^{-1}$ for 3.9V region, respectively. Undoubtedly, the D_{Na^+} of $\text{Na}_3\text{V}_2(\text{PO}_4)_3$ is slightly higher than that of $\text{Na}_3\text{V}_3(\text{PO}_4)_4$, which indicates that the redox couple of $\text{Na}_3\text{V}_2(\text{PO}_4)_3$ dominates the kinetics of sodium ions in composites, which is consistent with the previous conclusion.

Moreover, the dynamic characteristics of NVP/C and PVP-NVP were further explored by electrochemical impedance spectroscopy (EIS). Fig. 6d shows Nyquist plots for both NVP/C and PVP-NVP. It can be clearly observed that the high frequency region of the curve is semicircle and the low frequency region is a straight line, which corresponds to the transfer resistance (R_{ct}) and Warburg impedance (Z_w) of the material respectively. The equivalent circuit fitting curve is shown in the illustration in Fig. 6d. It can be intuitively observed from the high frequency region that the transfer resistance of PVP-NVP is 131.0 Ω , which is significantly lower than 750.1 Ω of NVP/C. This can be attributed to the highly ordered nitrogen-doped carbon layer transformed by PVP after sintering, which allows for the high-speed transport of electrons across the material surface. There is a linear relationship between Z' and $\omega^{-0.5}$ in the low frequency range, as shown in Fig. 6e. And the corresponding D_{Na^+} can be calculated by the following equation:

$$Z_{te} = R_s + R_{ct} + \sigma \omega^{-0.5} \quad (2)$$

$$D_{\text{Na}^+} = \frac{R^2 T^2}{2A^2 N^2 F^4 C_{\text{Na}} \sigma^2} \quad (3)$$

The related parameters and D_{Na^+} which calculated by Eq. (3) is shown in Table 2. Obviously, the Na^+ diffusion rate of PVP-NVP ($2.00 \times 10^{-15} \text{ cm}^2 \text{ s}^{-1}$) is significantly increased, almost three times that of NVP/C ($9.60 \times 10^{-16} \text{ cm}^2 \text{ s}^{-1}$), which was consistent with the CV curve. Therefore, it can be inferred that the two-phase composite possesses a larger migration channel, which can allow the rapid embedding and de-embedding of Na^+ in the crystal structure. This can mainly be attributed to the larger defects provided by the nitrogen-doped carbon cladding.

Furthermore, to further characterize the electrochemical kinetic properties of PVP-NVP, GITT experiments were performed. Specifically, the charging process is achieved by charging at 0.1C for 10 min and then holding it for 120 min to reach a steady state. The whole process is repeated until the voltage reaches the cut-off potential (4.1 V). And the same procedure was used for the GITT test during the discharge. Fig. 6f shows the GITT curves of PVP-NVP in detail. Both the charge and discharge curves exhibit two platforms at 3.4 V and 3.9 V, which can be attributed to $\text{Na}_3\text{V}_2(\text{PO}_4)_3$ and $\text{Na}_3\text{V}_3(\text{PO}_4)_4$, respectively. What's more, Fig. S5 are the single "potential versus time" and the linear "potential versus $\tau^{1/2}$ " in a charge or discharge period. Due to there is a linear

Table 2

The related parameters and D_{Na^+} of NVP/C and PVP-NVP deduced from EIS measurement.

Sample	R_{ct}/Ω	σ	$D_{\text{Na}^+}/\text{cm}^2 \text{ s}^{-1}$
NVP/C	750.1	217.85	9.60×10^{-16}
PVP-NVP	131.0	150.93	2.00×10^{-15}

relationship between potential and $\tau^{1/2}$ after fitting, D_{Na^+} can be calculated according to equation (5), which is simplified from equation (4) and m , V , M and A represent weight, molar volume, molar mass of active material and area of electrode, respectively.

$$D_{\text{Na}^+} = \frac{4}{\pi} \left(\frac{mV_m}{MA} \right) \left(\frac{\Delta E_s}{\tau(dE_t/d\sqrt{\tau})} \right)^2 \left(\frac{\tau \leq L^2}{D_{\text{Na}^+}} \right) \quad (4)$$

$$D_{\text{Na}^+} = \frac{4}{\pi} \left(\frac{mV_m}{MA} \right)^2 \left(\frac{\Delta E_s}{\Delta E_t} \right)^2 \quad (5)$$

Correspondingly, the D_{Na^+} of PVP-NVP with different potential is shown in Fig. 6g. The D_{Na^+} of PVP-NVP is 10^{-8} magnitude order, demonstrating the outstanding Na^+ storage process of it. Besides, D_{Na^+} has an obvious downward trend near 3.4 V and 3.9 V during the charging and discharging of PVP-NVP, which can be attributed to the phase transfer of $\text{Na}_3\text{V}_2(\text{PO}_4)_3$ and $\text{Na}_3\text{V}_3(\text{PO}_4)_4$ at their charging and discharging platforms, respectively. This is consistent with the results of the above material composition analysis and electrochemical performance test.

For further investigating the mechanism of the two charging and discharging platforms which located at 3.4 V and 3.9 V, ex-situ XPS test was performed during the charge/discharge process (Fig. 7a). There are eight different potentials shown in Fig. 7a to represent the change of V valence state during the charging and discharging process. Specifically, as exhibited in Fig. 7b, V^{3+} gradually changes to V^{4+} during charging and back to V^{3+} in discharge process. When the voltage reaches at 3.4 V, there are +3 and +4 valence states of V, which indicates that the reciprocal transition from $\text{Na}_3\text{V}_2(\text{PO}_4)_3$ to $\text{Na}_3\text{V}_3(\text{PO}_4)_4$ phases, corresponding to the phenomenon of low diffusion coefficient of Na^+ in GITT. When the voltage located at 3.9 V, it corresponds to the charging and discharging platform of the $\text{Na}_3\text{V}_3(\text{PO}_4)_4$. V has two valence states, namely +3 and +4, which correspond to the removal of one Na^+ in the $\text{Na}_3\text{V}_3(\text{PO}_4)_4$, which corresponds to the low diffusion coefficient of sodium ion at 3.9 V in the GITT curve. Furthermore, Fig. 7c shows the valence change trend of element V which charged from 3.4 V to 4.0 V and discharged back to 3.4 V. Obviously, the valence state of V element is the same at equal potential, which indicates that the change law of valence state of V element in the process of charging and discharging is reversible.

Fig. 8 displays the formation of $\text{Na}_3\text{V}_2(\text{PO}_4)_3$ and $\text{Na}_3\text{V}_3(\text{PO}_4)_4$ which induced by PVP as a structural guiding agent [42,43]. PVP controls phase formation in precursor solution by two mechanisms. On the one hand, during the growth of $\text{Na}_3\text{V}_3(\text{PO}_4)_4$, PVP mainly acts as a one-dimensional cross-linked organic template to induce precursor growth into $\text{Na}_3\text{V}_3(\text{PO}_4)_4$ crystals with layered structures. Specifically, the amide structure in PVP allows the precursor to be attracted to the surface of the cross-linked template through coordination and induces the formation of layered $\text{Na}_3\text{V}_3(\text{PO}_4)_4$ crystals. On the other hand, the O and N atoms in PVP molecules can provide lone electron pairs to form coordination complexes in aqueous solution. Therefore, PVP acts as a chelating agent in the precursor solution to increase steric hindrance between precursor particles to generate $\text{Na}_3\text{V}_2(\text{PO}_4)_3$ particles [44]. In addition, nitrogen atoms on five-membered rings in PVP are reductive. Furthermore, PVP also acts as a reducing agent in the precursor solution to make the reduction reaction from V^{5+} to V^{3+} faster and fuller, simplifying the preparation steps [45,46]. Furthermore, nitrogen doped

Table 1

The D_{Na^+} for NVP/C and PVP-NVP calculated by Randles-Sevcik method.

Sample	Slope		$D_{\text{Na}^+}/\text{cm}^2 \text{ s}^{-1}$	
	Charge	Discharge	Charge	Discharge
NVP/C	0.02690	-0.01230	2.58×10^{-11}	5.39×10^{-12}
PVP-NVP (~3.4V)	0.03003	-0.01117	3.21×10^{-11}	4.66×10^{-12}
PVP-NVP (~3.9V)	0.02202	-0.01863	1.73×10^{-11}	1.24×10^{-11}

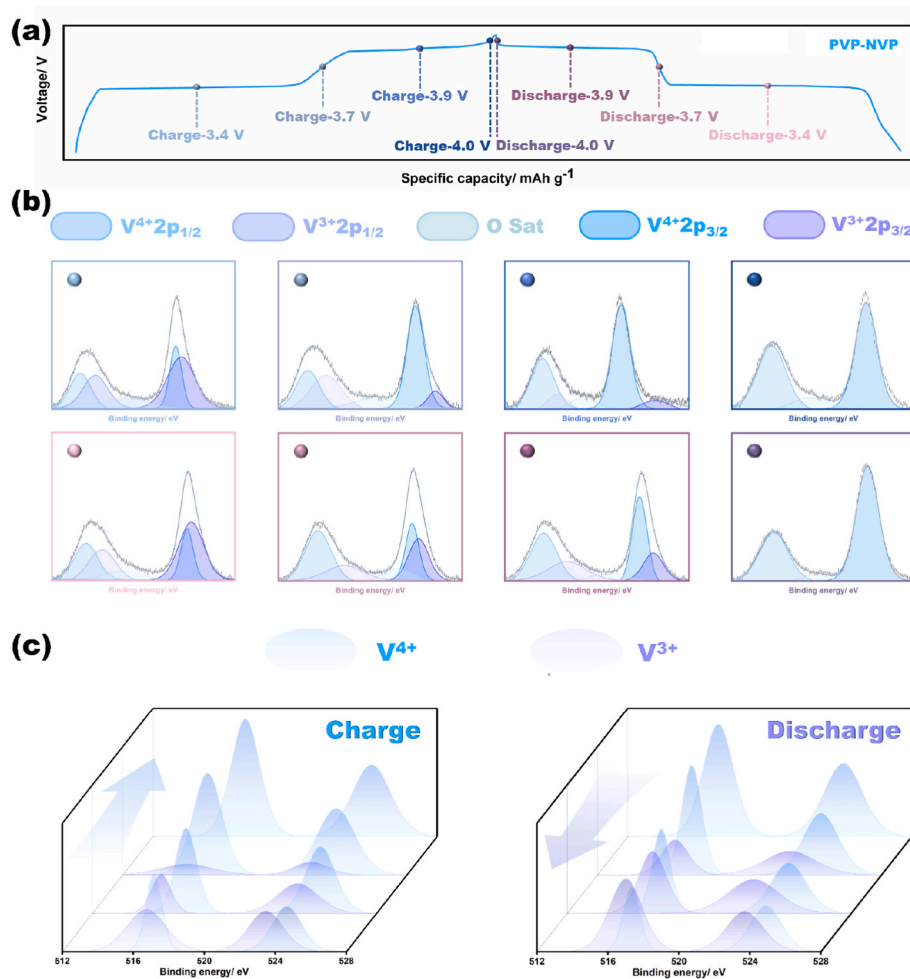


Fig. 7. (a) GCD curves of PVP-NVP. (b) Ex-situ XPS results of V detected at different charge–discharge state corresponding to the points in (a). (c) Valency trend diagram of V element of PVP-NVP in charge and discharge.

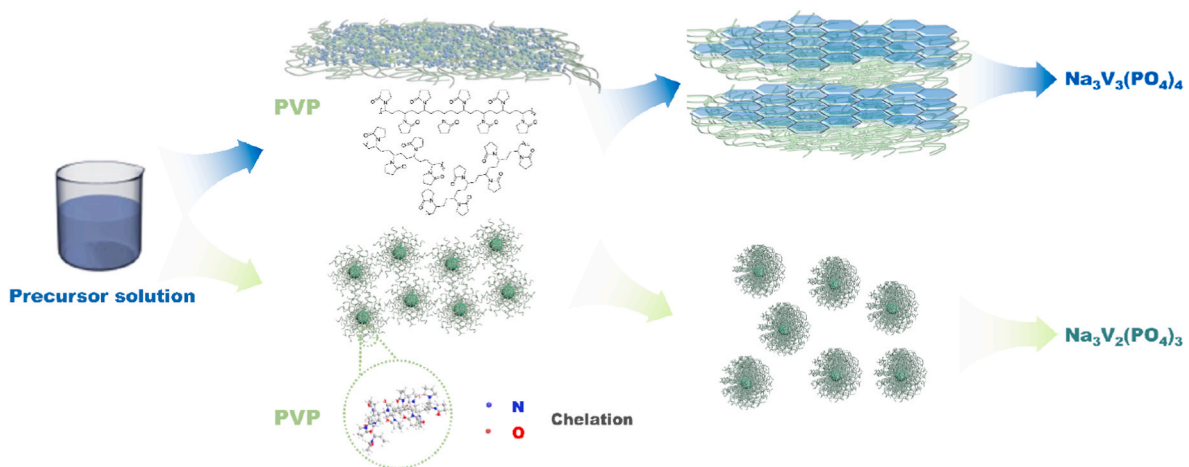


Fig. 8. Schematic diagram of PVP as a structural guide to generate two phase composite materials.

carbon coating formed by PVP after sintering has more defects. This carbon layer not only allows the high-speed transmission of electrons on its surface, but also allows the rapid diffusion of sodium ions. Moreover, PVP, as the main source of carbon coating, will form clusters and free radicals during thermal decomposition, thus promoting the orderly arrangement of some carbon nanoparticles and forming highly

graphitized carbon. As a result, the carbon coating can both transport electrons at high speeds and allow sodium ions to spread quickly.

4. Conclusions

In summary, Na₃V₂(PO₄)₃/Na₃V₃(PO₄)₄ two-phase composites were

successfully synthesized by a facile sol-gel method for the first time by adding PVP. PVP acts as a structural guide, reducing agent and dispersant in precursor solutions, inducing the formation of two-phase substances. The composite material submits an obvious high voltage platform ($\sim 3.9\text{V}$), which can be attributed to the formation of $\text{Na}_3\text{V}_3(\text{PO}_4)_4$. Furthermore, PVP decomposed into clusters and free radicals during carbonization to form uniform carbon nanoparticles with relatively ordered structure, resulting in highly graphitized carbon coatings and disordered carbon cladding. This blended cathode material delivers excellent electrochemical properties. PVP-NVP delivers a high discharge capacity of 119.17 mAh g^{-1} at 0.1C , much higher than that of bare NVP/C electrode (67.40 mAh g^{-1}). Besides, PVP-NVP possess a reversible specific capacity of $102.60\text{ mA h g}^{-1}$ after 200 cycles at 0.1C . When the current density increases to 10C , PVP-NVP can still provide discharge capacity of 86.80 mAh g^{-1} . What's more, benefiting from the high voltage profiles which come from $\text{Na}_3\text{V}_3(\text{PO}_4)_4$, PVP-NVP generates an enhanced initial energy density of 471.19 Wh kg^{-1} . It submits an average energy density of 408.12 Wh kg^{-1} after 200 cycles at 0.1C and exhibits superior energy density at different current densities. The novel strategy of PVP in situ induced formation of $\text{Na}_3\text{V}_2(\text{PO}_4)_3/\text{Na}_3\text{V}_3(\text{PO}_4)_4$ two-phase composites is a prospective route for cathode materials with high performance in SIBs.

CRediT authorship contribution statement

Jiahao Li: Investigation, Writing – original draft. **YanJun Chen:** Formal analysis, Writing – review & editing, Funding acquisition. **Hongen Shi:** Conceptualization, Validation. **Tao Zhou:** Conceptualization. **Zhen Tian:** Resources, Conceptualization, Supervision. **Yanzhong Wang:** Resources, Supervision. **Li Guo:** Resources, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jpowsour.2023.232802>.

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